Happy Supplies

a. State any assumptions you make about the data and the attributes shown in this table.

→ Assumptions are that Part Number is a number for a specific instrument that could be repeated. Since one type of instrument could be ordered several times.

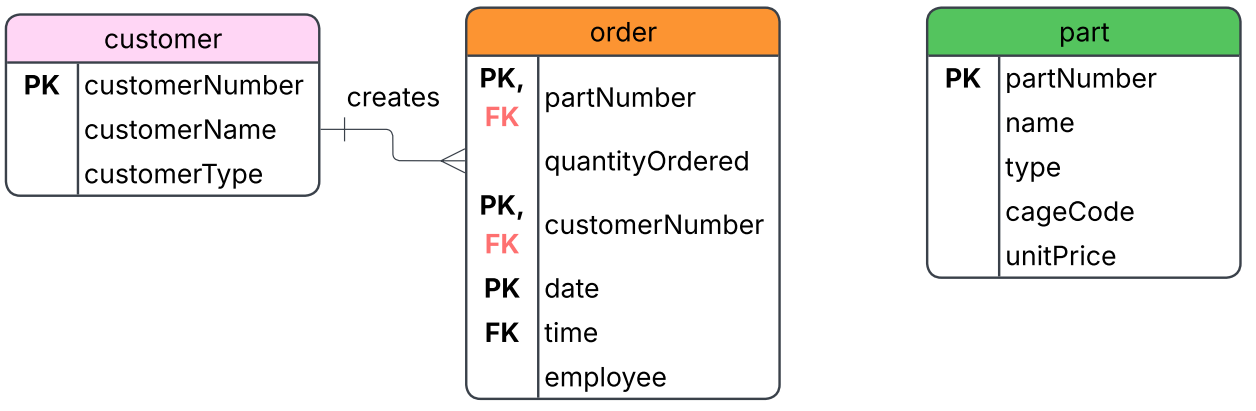
I created a big table with all the given information. It is 1NF because all the values are atomic, and there are no repeating groups. The composite Primary Key is partNumber + customerNumber + Date + Time.

partNumber	name	type	cageCode	quantityOrdered	unitPrice	customerNumber	customerName	customerType	date	time	employee
PK						PK			PK	PK	

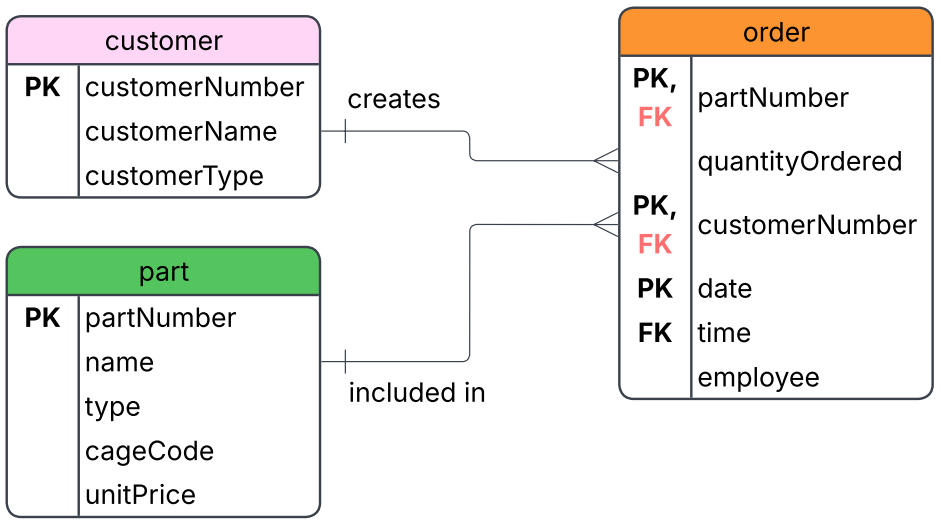
b. Describe and illustrate the process of normalizing the table to 3NF relations.

Step 1: turning into 2NF: 1NF + no partial dependencies.
We remove the partial dependency of :

- ***name, type, cageCode, unitPrice*** -> dependent on ***partNumber***
- ***customerName, customerType*** -> dependent on **customerNumber**



Step 2: turning into 3NF: 2NF+ no transitive dependencies.
There are no transitive dependencies. Therefore the database is in 3NF.



Mental Health Corporation

a. State any assumptions you make about the data and the attributes shown in this table.

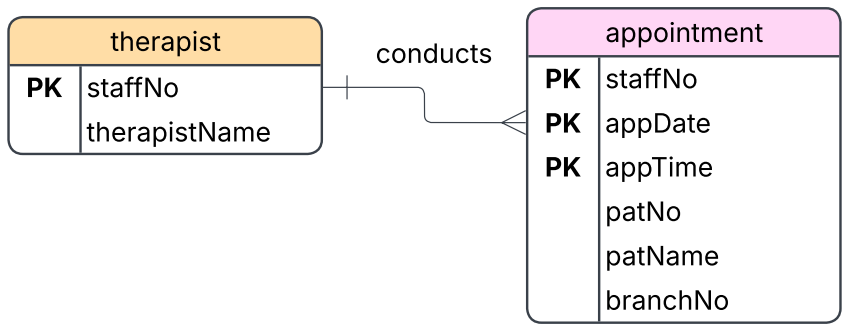
→ The table is not in 1NF Because, there is an attribute - appointments, which is not atomic. It has two sub attributes: Date and Time.

b. Describe and illustrate the process of normalizing the table to 3NF relations.

Step 1: turning into 1NF. We need the table to have primary key and to have atomic values.

appointment	
PK	staffNo
PK	appDate
PK	appTime
	therapistName
	patNo
	patName
	branchNo

Step 2: turning into 2NF: 1NF+ no partial dependencies.
We remove the partial dependency of ***therapistName*** on ***staffNo*** only.



Step 3: turning into 3NF: 2NF+ no transitive dependencies.
We remove the transitive dependency of ***patName*** on ***patNo***.

